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## 教育背景

**浙江大学，竺可桢学院** 2023/09 - 2027/06（预期）

计算机科学与技术专业，*23* 级本科 GPA: 3.97/4.00

## 论文成果

**DraftKV: Efficient Long-Term Memory via Attention-Guided KV Retrieval for Video Generation** 2026/10

**In Progress Project**

**Draft Attention v2: Fast Video Diffusion with Low-Resolution-Guided Mixed-Precision Attention** 2026/09

**Under Review**

*Rui Ding, Haopeng Li, Weize Ma, Yufa Zhou, Yitong Li,* ***Xiaoling Zhou****, Jiashuo Cao, Liyang Li, Hua Geng, Jiuxiang Gu, Jun Lin, Chunhua Shen, Enze Xie, Xuan Shen*

**MosaicAttention: Hierarchical Block-Sparse Attention for Multi-View Geometry Transformers** 2026/09

**Under Review**

*Zhizhen Pan, Rui Ding, Weize Ma,* ***Xiaoling Zhou****, Jun Lin, Chunhua Shen, Xuan Shen*

**Mind Dreamer: Untethering Imagination via Active Latent Intervention on Latent Manifolds arXiv** 2026/05

**ICML26 (CCF-A) Accepted**

*Shaojun Xu,* ***Xiaoling Zhou****, Yihan Lin, Yapeng Meng, Xinglong Ji, Luping Shi, Rong Zhao*

## 科研经历

**DraftKV: 面向长时序视频生成的注意力引导 KV 记忆检索** 2026/09 - 2026/10

- 发现长时序视频世界模型存在**长期记忆与推理开销的矛盾**：滑动窗口容易遗忘远距离关键场景，而保留完整历史 KV 又会持续增加显存、数据搬运与注意力计算开销。
- 提出 **DraftKV**，使用**低分辨率注意力**引导的历史 KV 检索，从 CPU 全精度历史 KV 检索**局部时序记忆与远程显著记忆**。显著提升视频生成质量和长历史一致性，在 SolarWM-H3 33B 上降低 **35.4%**GPU 峰值显存。
- 提出 **Streaming FlashAttention** 机制，实现 H2D 和注意力计算时间高度重叠。在 LongLive 1.0 的 60s 视频生成中实现 **1.21 倍端到端加速**，**Fetch+Attention 流水线**在 RTX4090/5090 上分别获得 **1.91x /2.51x** 加速。

**DraftAttention2: 基于低分辨率注意力引导的混合精度视频生成加速** 2026/07 - 2026/09

- 发现长视频生成中的**效率–质量矛盾**：纯稀疏注意力会直接丢失信息，在少步扩散中尤其难以恢复；全局低比特量化则会对关键视觉交互引入额外误差。
- 提出 **DraftAttention2**，利用平均/最大双路池化构建**低分辨率注意力重要性估计**，并据此将关键、次关键与低贡献交互分别分配至 16-bit、8/4-bit 和稀疏计算；进一步通过**注意力输出误差上界分析**刻画混合精度相对纯稀疏计算的优势。
- 设计**融合量化、数据重排与跨精度 Online Softmax 的 CUDA Kernel**，在单 Kernel 内完成多精度注意力计算与累加，减少中间结果写回与冗余访存，将算法层面的计算节省转化为实际推理加速。
- 在 HunyuanVideo-13B 与 MiniMax-H3 上分别实现最高 **2.51 倍/2.60 倍** DiT 推理加速；在 8-step 视频生成中实现 **1.87 倍**加速，同时保持接近 Dense Attention 的生成质量。

**MosaicAttention: 面向多视角几何 Transformer 的稀疏注意力加速** 2026/07 - 2026/09

- 发现多视角几何 Transformer 中的全局注意力开销随视角数量平方增长，而现有稀疏方法往往忽略**图像二维空间结构**，难以准确保留关键跨视角交互。
- 提出 **MosaicAttention**，通过二维平均/最大双路池化估计视觉区域重要性，并结合**层级化稀疏路由**按网络层动态分配计算预算，在保留关键几何信息的同时减少冗余注意力计算；进一步结合混合精度计算提升执行效率。
- 在 CO3Dv2 等多视角几何任务上，在保持或提升推理精度的同时，实现 128 视角输入下最高 **3.12 倍**端到端推理加速。

**Mind Dreamer: 通过主动干预打破世界模型想象的历史束缚** 2025/10 - 2026/05

- 发现世界模型想象存在**历史束缚（Historical Tethering）**：想象通常从历史状态出发，导致策略难以探索**未访问但高学习价值的区域**，限制稀疏奖励任务中的样本效率。
- 提出 **Mind Dreamer**，主动生成具有高学习价值的潜在状态作为新的想象起点，并结合**任务价值与不确定性**评估，引导策略跳出历史轨迹、探索更有价值的区域。
- 在 DeepMind Control Suite 视觉控制任务上，相比 DreamerV3 实现平均 **1.67 倍**、最高 **8.8 倍**样本效率提升，并在多个稀疏奖励任务上取得更高最终回报。

**Beyond Token Supervision: 通过语义锚点监督大模型隐式推理思维链** 2025/05 – 2025/09

- 发现 LLM 隐式推理缺乏显式中间监督，容易在连续隐空间推理过程中产生**语义漂移**，导致后续推理偏离正确思路。
- 提出**语义锚点监督机制**，利用 SAE 从显式 CoT 模型的中间隐状态中提取语义特征，并将其作为监督信号约束隐式推理过程，使中间状态保持与正确推理语义一致。
- 在 GSM8K 上较 SOTA 方法 COCONUT 实现 **11.7% 准确率提升**，有效缓解隐式推理中的语义分布偏移。

## 荣誉奖项

**国家奖学金**（本科生，排名前 *0.6%*） 2024/10

**浙江大学一等奖学金** *(连续获得 2 次，前 10%)* 2024/10, 2025/10

**竺可桢学院领航奖学金** 2024/11

**浙江省大学生 (理论) 物理竞赛二等奖** 2024/12

## 专业技能

**编程语言:** Python, C/C++, CUDA
**框架与工具:** PyTorch, MuJoCo, Git, LaTeX

# Xiaoling Zhou

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## Education

**Zhejiang University, Chu Kochen Honors College** Sept. 2023 – June 2027 (Expected)
*B.Eng. in Computer Science and Technology* GPA: 3.97/4.00

## Publications & Manuscripts

**DraftKV: Efficient Long-Term Memory via Attention-Guided KV Retrieval for Video Generation** Oct. 2026
**In Progress Project**

**Draft Attention v2: Fast Video Diffusion with Low-Resolution-Guided Mixed-Precision Attention** Sept. 2026
**Under Review**
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**MosaicAttention: Hierarchical Block-Sparse Attention for Multi-View Geometry Transformers** Sept. 2026
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**Mind Dreamer: Untethering Imagination via Active Causal Intervention on Latent Manifolds** **arXiv** May 2026
**In Proceedings of ICML 2026 (CCF-A)**
*Shaojun Xu, **Xiaoling Zhou**, Yihan Lin, Yapeng Meng, Xinglong Ji, Luping Shi, Rong Zhao*

## Research Experience

**DraftKV: Attention-Guided KV Retrieval for Long-Horizon Video Generation** Sept. 2026 – Oct. 2026
- Identified a **long-term memory–cost trade-off**: sliding windows forget distant scenes, while full-history KV incurs growing GPU memory, transfer, and attention costs.
- Developed **DraftKV**, extending DraftAttention2's **low-resolution attention scoring** to historical KV retrieval; retrieves local and salient long-range memories from full-precision CPU-resident KV, improving quality while reducing peak GPU memory by **35.4%** on SolarWM-H3 33B.
- Designed **Streaming FlashAttention** to overlap H2D KV transfer with attention, achieving **1.21× end-to-end speedup** for 60-s LongLive 1.0 generation and **1.91×/2.51×** Fetch+Attention speedups on RTX 4090/5090.

**DraftAttention2: Low-Resolution-Guided Mixed-Precision Attention for Video Generation** July 2026 – Sept. 2026
- Identified an **efficiency–quality trade-off**: sparse attention irreversibly drops information in few-step diffusion, while global low-bit quantization perturbs critical visual interactions.
- Developed **DraftAttention2**, using average/max dual-path pooling for **low-resolution attention importance estimation** and routing interactions to 16-bit, 8/4-bit, or sparse computation; derived an **attention-output error bound** showing the advantage over pure sparsity.
- Designed a **fused CUDA kernel** for quantization, data reordering, and cross-precision Online Softmax, reducing intermediate writes and redundant memory traffic.
- Achieved up to **2.51×/2.60× DiT inference speedups** on HunyuanVideo-13B/MiniMax-H3 and **1.87×** on 8-step generation while maintaining near-dense quality.

**MosaicAttention: Sparse Attention for Multi-View Geometry Transformers** July 2026 – Sept. 2026
- Identified a limitation of existing sparse attention for multi-view geometry: global attention scales quadratically with views, while conventional sparsification ignores **2D image structure** and critical cross-view interactions.
- Developed **MosaicAttention**, combining 2D average/max pooling with **hierarchical sparse routing** to allocate computation across layers, preserve key geometric interactions, and remove redundant attention; further incorporated mixed-precision execution.
- Achieved up to **3.12× end-to-end inference speedup** at 128 views while matching or improving accuracy on CO3Dv2 and other multi-view geometry tasks.

**Mind Dreamer: Untethering World-Model Imagination via Active Intervention** Oct. 2025 – May 2026
- Identified **Historical Tethering** in model-based RL: imagination starts from observed states, limiting exploration of **unvisited high-value regions** and sample efficiency in sparse-reward tasks.
- Developed **Mind Dreamer**, synthesizing high-value latent states as new imagination starting points and jointly scoring **task value and uncertainty** to guide exploration beyond historical trajectories.
- Achieved **1.67× average** and up to **8.8× sample-efficiency gains** over DreamerV3 on DeepMind Control Suite, with higher final returns on multiple sparse-reward tasks.

**Beyond Token Supervision: Semantic Anchors for Latent Reasoning in LLMs** May 2025 – Sept. 2025
- Identified **semantic drift** in latent reasoning: without intermediate supervision, latent states can deviate from the intended reasoning trajectory.
- Developed **semantic-anchor supervision**, using sparse autoencoders (SAEs) to extract semantic features from an explicit-CoT teacher and align intermediate latent states with correct reasoning semantics.
- Improved GSM8K accuracy by **11.7%** over the SOTA latent-reasoning baseline COCONUT, substantially reducing semantic drift.

## Awards

**National Scholarship** *(Top 0.6%)* Oct. 2024
**Zhejiang University First-Class Scholarship** *(Twice, Top 10%)* Oct. 2024, Oct. 2025
**Chu Kochen Honors College Leading Scholarship** Nov. 2024
**Zhejiang Provincial Undergraduate Physics Competition, Second Prize** Dec. 2024

## Skills

**Programming:** Python, C/C++, CUDA
**Frameworks & Tools:** PyTorch, MuJoCo, Git, LaTeX